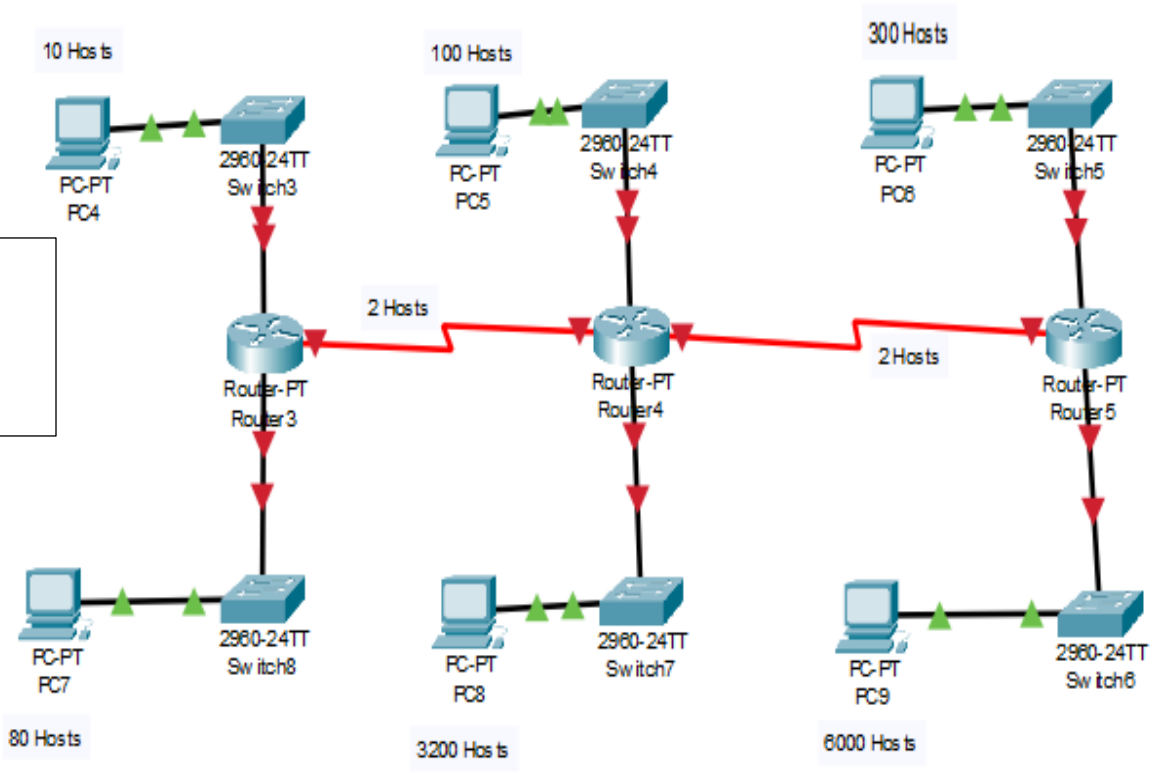
Implementation of different types of LAN topology with IP addressing using PT

Objectives:

- Design the following topology in Packet Tracer.
- Subnet each of the network with the IP and complete the following table.
- Assign the appropriate IP into each of the network.
- Add one more PC to each of router and assign IP.
- Test connectivity between PC in a router.

Address Space

192.168.1.0 / 20



First subnet (6000 hosts)

Required hosts = 6000 ≈ 8192 = 2^13 ,Prefix portion = 32 -13 = 19

Subnet mask = 11111111. 11111111.11100000. 00000000 =255.255.224.0

N.A= 192.168.1.0 /19

B.A= 192.168.000000001.00000000
192.168.00011111.11111111
192.168.31.255 /19

Second subnet (3200 hosts)

Required hosts 3200 ≈ 4096 = 2^12 ,Prefix portion = 32 -12 = 20

Subnet mask = 11111111. 11111111.11110000. 00000000 =255.255.240.0

N.A = 192.168.32.0 /20

B.A= 192.168.00100000.00000000
192.168.00101111.11111111
192.168.47.255/20

3rd subnet (300 hosts)

Required hosts 300 ≈ 512 = 2^9 ,Prefix portion = 32 -9 = 23

Subnet mask = 11111111. 11111111.11111110. 00000000 =255.255.254.0

N.A = 192.168.48.0 /23

B.A= 192.168.00110000.00000000
192.168.00110001.11111111
192.168.49.255/23

4th subnet (100 hosts)

Required hosts 100 ≈ 128 = 2^7 ,Prefix portion = 32 -7 = 25

Subnet mask = 11111111. 11111111.11111111. 10000000 =255.255.255.128

N.A = 192.168.50.0 /25

B.A= 192.168.00110010.00000000
192.168.00110010.01111111
192.168.50.127/25

5th subnet (80 hosts)

Required hosts 80 ≈ 128 = 2^7 ,Prefix portion = 32 -7 = 25

Subnet mask = 11111111. 11111111.11111111. 10000000 =255.255.255.128

N.A= 192.168.50.128 /25 ---------- B.A= 192.168.50.255/25

6th subnet (2 hosts)

Required hosts 2 ≈ 4 = 2^2 hosts ,Prefix portion = 32 -2 = 30

Subnet mask= 11111111. 11111111.11111111. 11111100 =255.255.255.252

N.A= 192.168.51.0 /30 ---------- B.A= 192.168.51.3/30

7th subnet (2 hosts)

Required hosts 2 ≈ 4 = 2^2 hosts ,Prefix portion = 32 -2 = 30

Subnet mask= 11111111. 11111111.11111111. 11111100 =255.255.255.252

N.A= 192.168.51.4 /30 ---------- B.A= 192.168.51.7/30

Subnet Table

Subnet	Assignable Adress Range	Subnet Mask	Subnet Address	Broadcast Address
First subnet (6000 hosts)	192.168.1.1 /19 - 192.168.31.254 /19	255.255.224.0	192.168.1.0 /19	192.168.31.255 /19
Second subnet (3200 hosts)	192.168.32.1 /20- 192.168.47.254/20	255.255.240.0	192.168.32.0 /20	192.168.47.255/20
3rd subnet (300 hosts)	192.168.48.1 /23 - 192.168.49.254 /23	255.255.254.0	192.168.48.0 /23	192.168.49.255/23
4th subnet (100 hosts)	192.168.50.1 /25 - 192.168.50.126/25	255.255.255.128	192.168.50.0 /25	192.168.50.127/25
5th subnet (80 hosts)	192.168.50.129 /25- 192.168.50.254/25	255.255.255.128	192.168.50.128 /25	192.168.50.255/25
6th subnet (2 hosts)	192.168.51.1 /30 – 192.168.51.2/30	255.255.255.252	192.168.51.0 /30	192.168.51.3/30
7th subnet (2 hosts)	192.168.51.5 /30 – 192.168.51.6/30	255.255.255.252	192.168.51.4 /30	192.168.51.7/30

Addressing table

Device	Interface	IP address	Subnet mask	Default Gateway
R3	Fast ethernet 0/0			N/A
	Serial 2/0			
	Serial 3/0			
R4	Fast ethernet 0/0			
	Serial 2/0			
R5	Fast ethernet 0/0			
	Serial 3/0			
PC4	N/A			
PC5	N/A			
PC6	N/A			
PC7	N/A			
PC8	N/A			
PC9	N/A			